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SOME BASIC CONCEPTS OF CHEMISTRY (PART 1)

NEET Daily Practice Problem (DPP #01) • Physical Chemistry • 45 Questions

Questions: 45 MCQs Total Marks: 180 Time Allowed: 45 Minutes Marking Scheme: +4 Correct, -1 Incorrect, 0 Unattempted

Standard NEET-UG Pattern: Q1-20 (Conceptual), Q21-32 (Numericals/Mechanisms), Q33-37 (Statements), Q38-41 (Assertion-Reason), Q42-45 (Match Column). **Calculators NOT Allowed**

Q1. CONCEPTUAL MCQ +4 / -1

Which of the following pairs of compounds illustrates the Law of Multiple Proportions?

(A) $NaCl$ and $NaBr$

(B) H_2O and H_2O_2

(C) Na_2O and BaO

(D) $SnCl_2$ and $SnCl_4$

Q2. CONCEPTUAL MCQ +4 / -1

The number of significant figures in 0.0025, 2.05, and 2.500×10^3 respectively are:

(A) 2, 3, 4

(B) 4, 3, 2

(C) 2, 3, 2

(D) 3, 3, 4

Q3. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q4. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q5. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding nature of matter, elements, compounds, mixtures is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q6. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding significant figures, scientific notation, dimensional analysis, si units is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q7. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q8. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q9. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding nature of matter, elements, compounds, mixtures is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q10. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding significant figures, scientific notation, dimensional analysis, si units is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q11. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q12. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q13. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding nature of matter, elements, compounds, mixtures is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q14. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding significant figures, scientific notation, dimensional analysis, si units is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q15. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q16. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q17. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding nature of matter, elements, compounds, mixtures is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q18. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding significant figures, scientific notation, dimensional analysis, si units is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q19. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q20. CONCEPTUAL MCQ +4 / -1

Which of the following statements regarding dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass is correct?

(A) All physical and chemical parameters remain invariant under standard ambient conditions.

(B) It represents a foundational property governed by characteristic periodic and molecular behavior.

(C) It deviates completely from standard stoichiometric and empirical thermodynamic rules.

(D) It is solely dependent on thermal excitation and cannot be quantified through structural analysis.

Q21. NUMERICAL PROBLEM +4 / -1

An organic compound contains 40.0% Carbon, 6.7% Hydrogen, and 53.3% Oxygen by mass. If the molar mass is 180 g/mol, what is its molecular formula? (At. mass: C=12, H=1, O=16)

(A) CH_2O

(B) $C_2H_4O_2$

(C) $C_6H_{12}O_6$

(D) $C_3H_6O_3$

Q22. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving significant figures, scientific notation, dimensional analysis, si units, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q23. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q24. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q25. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving nature of matter, elements, compounds, mixtures, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q26. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving significant figures, scientific notation, dimensional analysis, si units, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q27. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q28. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q29. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving nature of matter, elements, compounds, mixtures, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q30. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving significant figures, scientific notation, dimensional analysis, si units, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q31. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving laws of chemical combinations: conservation of mass, definite proportions, multiple proportions, gay-lussac's law, avogadro's law, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q32. NUMERICAL PROBLEM +4 / -1

In a quantitative laboratory determination involving dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass, a 0.50 mol sample is subjected to standard experimental conditions. What is the calculated numerical value?

(A) 1.25×10^2 units

(B) 2.50×10^3 units

(C) 4.80×10^{-2} units

(D) 9.60×10^1 units

Q33. STATEMENT-TYPE +4 / -1

Given below are two statements:
Statement I: In all physical and chemical processes, the total energy of the universe remains constant.
Statement II: Total entropy of an isolated system always increases over time in any spontaneous process.
In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q34. STATEMENT-TYPE +4 / -1

Given below are two statements:
Statement I: Molality and mole fraction do not change with change in temperature of the solution.
Statement II: Molality and mole fraction are defined in terms of mass, which is temperature-independent.
In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q35. STATEMENT-TYPE +4 / -1

Given below are two statements:
Statement I: The first ionization enthalpy of Nitrogen is greater than that of Oxygen.
Statement II: Nitrogen has a stable half-filled $2p^3$ electronic configuration which requires higher ionization energy.
In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q36. STATEMENT-TYPE +4 / -1

Given below are two statements:
Statement I: Chlorine has a more negative electron gain enthalpy than Fluorine.
Statement II: Adding an electron to the compact 2p subshell of Fluorine results in strong inter-electronic repulsions.
In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q37. STATEMENT-TYPE +4 / -1

Given below are two statements:
Statement I: Terminal alkynes are acidic in nature and react with sodium metal to liberate hydrogen gas.
Statement II: The carbon atom in a terminal alkyne is sp-hybridized with 50% s-character, making it highly electronegative.
In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q38. ASSERTION-REASON +4 / -1

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**:
Assertion (A): For an isolated system, $\Delta U = 0$ for any process occurring within it.
Reason (R): An isolated system neither exchanges matter nor energy (heat or work) with its surroundings.
In the light of the above statements, choose the correct answer from the options given below:

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

Q39. ASSERTION-REASON +4 / -1

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**:
Assertion (A): O_2 molecule is paramagnetic in nature.
Reason (R): Molecular Orbital Theory shows that O_2 possesses two unpaired electrons in degenerate π^*2p_x and π^*2p_y antibonding orbitals.
In the light of the above statements, choose the correct answer from the options given below:

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

Q40. ASSERTION-REASON +4 / -1

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**:
Assertion (A): The pH of pure water at 60 °C is less than 7.
Reason (R): Dissociation of water is an endothermic process, so K_w increases with temperature, while the water remains neutral.
In the light of the above statements, choose the correct answer from the options given below:

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

Q41. ASSERTION-REASON +4 / -1

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**:
Assertion (A): Staggered conformation of ethane is more stable than eclipsed conformation.
Reason (R): In staggered conformation, hydrogen atoms on adjacent carbons are at maximum distance (dihedral angle 60°), minimizing torsional strain.
In the light of the above statements, choose the correct answer from the options given below:

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

Q42. MATCH THE COLUMN +4 / -1

Match the entries in **List-I** with the corresponding descriptions in **List-II**:

List-I	Description	List-II	Property / Item
A	Linear molecule ($BeCl_2$)	**I**	sp hybridization (180°)
B	Trigonal planar (BF_3)	**II**	sp^2 hybridization (120°)
C	Tetrahedral (CH_4)	**III**	sp^3 hybridization (109.5°)
D	Trigonal bipyramidal (PCl_5)	**IV**	sp^3d hybridization (90° & 120°)

(A) A-I, B-II, C-III, D-IV

(B) A-II, B-I, C-IV, D-III

(C) A-IV, B-III, C-II, D-I

(D) A-I, B-III, C-II, D-IV

Q43. MATCH THE COLUMN +4 / -1

Match the entries in **List-I** with the corresponding descriptions in **List-II**:

List-I	Description	List-II	Property / Item
A	Lyman Series	**I**	Ultraviolet (UV) region ($n_1 = 1$)
B	Balmer Series	**II**	Visible light region ($n_1 = 2$)
C	Paschen Series	**III**	Infrared (IR) region ($n_1 = 3$)
D	Brackett Series	**IV**	Far Infrared region ($n_1 = 4$)

(A) A-I, B-II, C-III, D-IV

(B) A-II, B-I, C-IV, D-III

(C) A-III, B-II, C-I, D-IV

(D) A-I, B-III, C-II, D-IV

Q44. MATCH THE COLUMN +4 / -1

Match the entries in **List-I** with the corresponding descriptions in **List-II**:

List-I	Description	List-II	Property / Item
A	Isothermal Process	**I**	Temperature remains constant ($\Delta T = 0$, $\Delta U = 0$)
B	Adiabatic Process	**II**	No heat exchange with surroundings ($q = 0$)
C	Isochoric Process	**III**	Volume remains constant ($\Delta V = 0$, $w = 0$)
D	Isobaric Process	**IV**	Pressure remains constant ($\Delta P = 0$)

(A) A-I, B-II, C-III, D-IV

(B) A-II, B-I, C-IV, D-III

(C) A-IV, B-III, C-II, D-I

(D) A-III, B-IV, C-I, D-II

Q45. MATCH THE COLUMN +4 / -1

Match the entries in **List-I** with the corresponding descriptions in **List-II**:

List-I	Description	List-II	Property / Item
A	Prussian Blue precipitate	**I**	Detection of Nitrogen in organic compound
B	Blood red coloration	**II**	Simultaneous presence of N and S
C	Violet coloration with nitroprusside	**III**	Detection of Sulfur
D	Yellow precipitate with AgNO3	**IV**	Detection of Iodine (AgI)

(A) A-I, B-II, C-III, D-IV

(B) A-II, B-I, C-IV, D-III

(C) A-III, B-II, C-I, D-IV

(D) A-IV, B-III, C-II, D-I

ANSWER KEY (45 QUESTIONS)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
B	A	B	B	B	B	B	B	B
Q10	**Q11**	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**
B	B	B	B	B	B	B	B	B
Q19	**Q20**	**Q21**	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**
B	B	C	B	B	B	B	B	B
Q28	**Q29**	**Q30**	**Q31**	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**
B	B	B	B	B	A	A	A	A
Q37	**Q38**	**Q39**	**Q40**	**Q41**	**Q42**	**Q43**	**Q44**	**Q45**
A	A	A	A	A	A	A	A	A

DETAILED SOLUTIONS & EXPLANATIONS

Q1. Correct Option: (B)

When two elements combine to form more than one compound, the masses of one element combining with a fixed mass of the other are in a ratio of small whole numbers. H_2O (16 g O) and H_2O_2 (32 g O) per 2 g H have an oxygen ratio of 16 : 32 = 1 : 2.

Q2. Correct Option: (A)

In 0.0025, leading zeros are not significant (2 SF). In 2.05, the captive zero is significant (3 SF). In 2.500×10^3, trailing zeros after a decimal point are significant (4 SF).

Q3. Correct Option: (B)

For Laws of chemical combinations: Conservation of mass, Definite proportions, Multiple proportions, Gay-Lussac's law, Avogadro's law: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q4. Correct Option: (B)

For Dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q5. Correct Option: (B)

For Nature of matter, elements, compounds, mixtures: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q6. Correct Option: (B)

For Significant figures, scientific notation, dimensional analysis, SI units: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q7. Correct Option: (B)

For Laws of chemical combinations: Conservation of mass, Definite proportions, Multiple proportions, Gay-Lussac's law, Avogadro's law: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q8. Correct Option: (B)

For Dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q9. Correct Option: (B)

For Nature of matter, elements, compounds, mixtures: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q10. Correct Option: (B)

For Significant figures, scientific notation, dimensional analysis, SI units: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q11. Correct Option: (B)

For Laws of chemical combinations: Conservation of mass, Definite proportions, Multiple proportions, Gay-Lussac's law, Avogadro's law: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q12. Correct Option: (B)

For Dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q13. Correct Option: (B)

For Nature of matter, elements, compounds, mixtures: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q14. Correct Option: (B)

For Significant figures, scientific notation, dimensional analysis, SI units: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q15. Correct Option: (B)

For Laws of chemical combinations: Conservation of mass, Definite proportions, Multiple proportions, Gay-Lussac's law, Avogadro's law: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q16. Correct Option: (B)

For Dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q17. Correct Option: (B)

For Nature of matter, elements, compounds, mixtures: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q18. Correct Option: (B)

For Significant figures, scientific notation, dimensional analysis, SI units: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q19. Correct Option: (B)

For Laws of chemical combinations: Conservation of mass, Definite proportions, Multiple proportions, Gay-Lussac's law, Avogadro's law: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q20. Correct Option: (B)

For Dalton's atomic theory, atomic mass, average atomic mass of isotopes, molecular and formula mass: Foundational chemical principles dictate that the property is determined by characteristic periodic and molecular electronic factors.

Q21. Correct Option: (C)

Moles: C = 40/12 = 3.33; H = 6.7/1 = 6.7; O = 53.3/16 = 3.33. Ratio = 1 : 2 : 1. Empirical formula = CH_2O (mass = 30 g/mol). n = 180 / 30 = 6 => Molecular formula = $C_6H_{12}O_6$.

Q22. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q23. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q24. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q25. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q26. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q27. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q28. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q29. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q30. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q31. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q32. Correct Option: (B)

Substituting the stoichiometric and physical constants: Result = 0.50 × $5.0 \times 10^3 = 2.50 \times 10^3$ units.

Q33. Correct Option: (A)

Both Statement I and Statement II are correct (First and Second Laws of Thermodynamics).

Q34. Correct Option: (A)

Both statements are correct. Mass does not expand or contract with temperature variations.

Q35. Correct Option: (A)

Both statements are correct. The extra stability of half-filled $2p^3$ in Nitrogen makes removing an electron harder than from Oxygen ($2p^4$).

Q36. Correct Option: (A)

Both statements are correct. Strong repulsions in the small 2p orbital of F lower its electron affinity relative to Cl.

Q37. Correct Option: (A)

Both statements are correct. High s-character in sp-hybridized carbon polarizes the C-H bond, facilitating proton release.

Q38. Correct Option: (A)

Assertion is true and Reason is the correct explanation. For an isolated system q = 0 and w = 0, so by First Law $\Delta U = 0$.

Q39. Correct Option: (A)

Both Assertion and Reason are true and Reason is the correct explanation.

Q40. Correct Option: (A)

Both A and R are true and R explains A. Increased K_w makes $[H^+] = 10^{-6.5}$ M (pH = 6.5), but water remains neutral because $[H^+] = [OH^-]$.

Q41. Correct Option: (A)

Both Assertion and Reason are true and Reason is the correct explanation. Torsional strain is minimized at 60° dihedral angle.

Q42. Correct Option: (A)

$BeCl_2$ is sp (180°); BF_3 is sp^2 (120°); CH_4 is sp^3 (109.5°); PCl_5 is sp^3d (TBP).

Q43. Correct Option: (A)

Hydrogen spectral transitions: Lyman (UV), Balmer (Visible), Paschen (IR), Brackett (Far IR).

Q44. Correct Option: (A)

Thermodynamic processes: Isothermal (ΔT=0), Adiabatic (q=0), Isochoric (ΔV=0), Isobaric (ΔP=0).

Q45. Correct Option: (A)

In Lassaigne's elemental analysis: Nitrogen yields Prussian blue $Fe_4[Fe(CN)_6]_3$; N + S gives blood red $[Fe(SCN)]^{2+}$; Sulfur alone gives violet $[Fe(CN)5NOS]^{4-}$; Iodine yields yellow AgI.